

Exercise for Chapter 12

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Chapter 12

12.1 To what temperature must MgCO_3 be heated in an atmosphere containing a partial pressure of CO_2 of 10^{-2} atm to cause the decomposition of the carbonate?

Chapter 12

Solution to 12.1

$$\Delta G^\circ = -RT \ln K \quad \text{MgO}_{(s)} + \text{CO}_{2(g)} = \text{MgCO}_{3(s)} \quad \text{Textbook P650, Table A.1}$$
$$\Delta G^\circ = -117600 + 170T = RT \ln p_{\text{CO}_2(\text{eq.}T)}$$

So, at $p_{\text{CO}_2(\text{eq})} = 10^{-2}$ atm,

$$-117600 + 170T = 8.3144 \cdot T \cdot \ln 10^{-2} = -38.29T$$

which gives $T = 565$ K.

Chapter 12

12.2 Using the standard Gibbs free energies of formation of NiO from solid Ni and liquid Ni, calculate the melting temperature, molar heat of melting, and the molar entropy of melting of nickel.

Chapter 12

Solution to 12.2

Textbook P650, Table A.1

For $2\text{Ni}_{(s)} + \text{O}_{2(g)} = 2\text{NiO}_{(s)} \quad \Delta G^\circ = -471200 + 172T \text{ J}$

And for $2\text{NiO}_{(s)} = \text{O}_{2(g)} + 2\text{Ni}_{(l)} \quad \Delta G^\circ = 506180 - 192.2T \text{ J}$

Adding gives

$$2\text{Ni}_{(s)} = 2\text{Ni}_{(l)} \quad \Delta G^\circ = 34980 - 20.2T \text{ J} = \Delta H^\circ - T\Delta S^\circ$$

So,

$$T_{m,\text{Ni}} = \frac{34980}{20.2} = 1731 \text{ K}$$

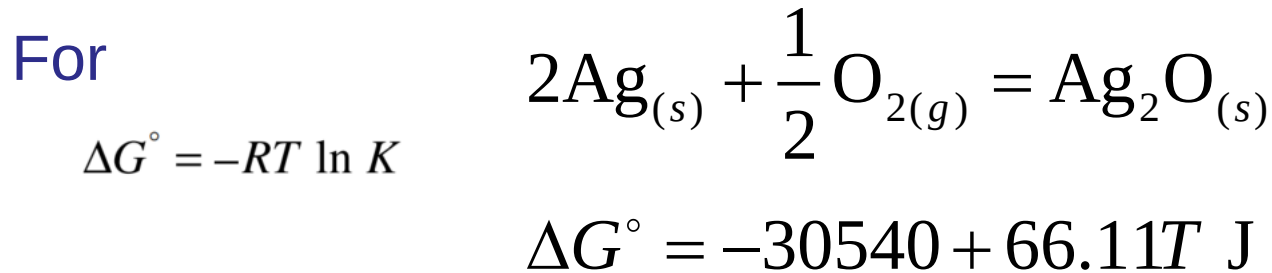
$$\Delta H_{m,\text{Ni}}^\circ = \frac{34980}{2} = 17490 \text{ J} \quad \Delta S_{m,\text{Ni}}^\circ = \frac{20.2}{2} = 10.1 \text{ J/K}$$

Chapter 12

12.3 Calculate the temperature at which pure Ag_2O decomposes to Ag metal and O_2 gas when heated in (a) pure oxygen at 1 atm pressure, and (b) air.

Chapter 12

Solution to 12.3



Textbook P650,
Table A.1

a. For $p_{\text{O}_2(\text{eq})} = 1 \text{ atm}$,

$$-30540 + 66.11T = RT \ln p_{\text{O}_2(\text{eq})}^{1/2} = 0$$

so $T = 462 \text{ K}$.

b. For $p_{\text{O}_2(\text{eq})} = 0.21 \text{ atm (in air)}$,

$$-30540 + 66.11T = RT \ln p_{\text{O}_2(\text{eq})}^{1/2} = \frac{1}{2} \times 8.3144 \times T \times \ln 0.21$$

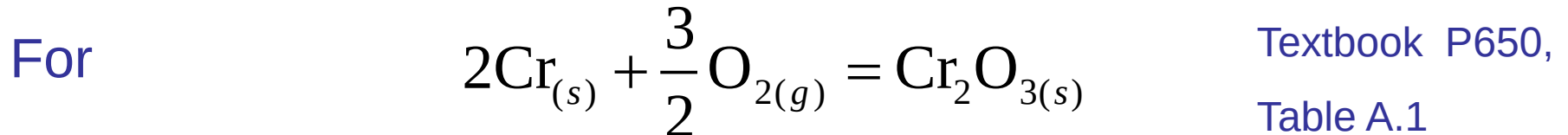
which gives $T = 421 \text{ K}$.

Chapter 12

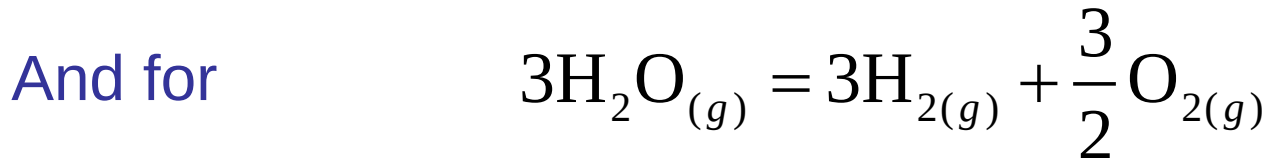
12.4 Determine the maximum pressure of water vapor in wet hydrogen at 1 atm pressure in which chromium can be heated without oxidation occurring at 1500 K. Is the oxidation of Cr by water vapor exothermic or endothermic?

Chapter 12

Solution to 12.4



$$\begin{aligned}\Delta G_{1500\text{ K}}^{\circ} &= -1,110,100 + 247.3T \text{ J} \\ &= -1,110,100 + 247.3 \times 1500 \text{ J} = -739150 \text{ J}\end{aligned}$$

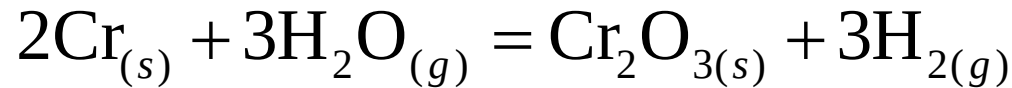


$$\begin{aligned}\Delta G_{1500\text{ K}}^{\circ} &= 3 \times (247,500 - 55.85T) \text{ J} \\ &= 3 \times (247,500 - 55.85 \times 1500) \text{ J} = 491175 \text{ J}\end{aligned}$$

Chapter 12

Solution to 12.4

Adding gives



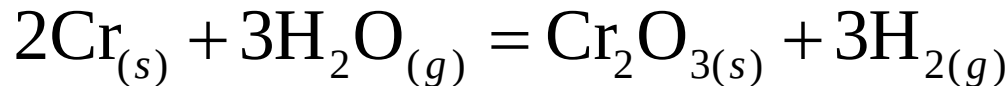
$$\Delta G_{1500\text{ K}}^{\circ} = -247,975\text{ J} = RT \ln \frac{p_{\text{H}_2\text{O}}^3}{p_{\text{H}_2}^3} = 1500 \times 8.3144 \times 3 \ln \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}$$

So

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 1.32 \times 10^{-3}$$

with $p_{\text{H}_2} = 1\text{ atm}$, $p_{\text{H}_2\text{O}(\text{max})} = 1.32 \times 10^{-3}\text{ atm}$

For



$\Delta H_{1500\text{ K}}^{\circ} = -371200\text{ J}$, and the reaction is exothermic.

Chapter 12

12.5 A mixture of argon gas and hydrogen gas at 1 atm total pressure is passed through a reaction vessel containing a mixture of liquid Sn and liquid SnCl_2 at 900 K. The composition of the gas leaving the vessel is 50% H_2 , 7% HCl , and 43% Ar. Has equilibrium been attained between the gas phase and the liquid phases in the vessel?

Chapter 12

Solution to 12.5

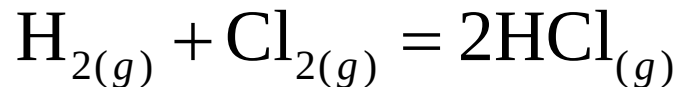
For



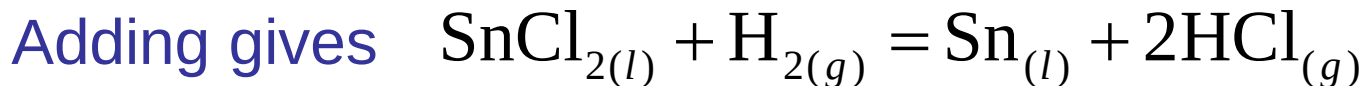
Textbook P650,
Table A.1

$$\Delta G_{900\text{ K}}^{\circ} = -(-333,000 + 118.4T) \text{ J} = 226440 \text{ J}$$

For



$$\Delta G_{900\text{ K}}^{\circ} = -188,200 - 12.80T \text{ J} = -199720 \text{ J}$$



$$\Delta G_{900\text{ K}}^{\circ} = 144,800 - 131.2T \text{ J} = 26720 \text{ J}$$

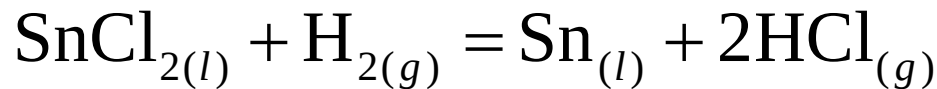
So $K = \exp\left(\frac{-\Delta G_{900\text{ K}}^{\circ}}{RT}\right) = \exp\left(\frac{-26720}{8.3144 \times 900}\right) \text{ J} = 0.028 = \frac{p_{\text{HCl}}^2}{p_{\text{H}_2}}$

Chapter 12

Solution to 12.5

For the final gas, $\frac{p_{\text{HCl}}^2}{p_{\text{H}_2}} = \frac{0.07^2}{0.5} = 0.0098$, which being less than K , indicates that equilibrium has not been attained.

The gas entering contains $0.5_{(\text{leave})} + 0.035_{(\text{reaction})} = 0.535$ mols of H_2 and 0.43 mols of Ar.



before 0.535

after 0.535-x $2x$ $n_{\text{total}} = 0.535 + x + 0.43$
 $= 0.965 + x$ mols

So $K = 0.028 = \frac{(2x)^2}{(0.965 + x)(0.535 - x)}$, which gives $x = 0.0584$.

Chapter 12

Solution to 12.5

For the equilibrium state, the gas composition is:

$$\text{HCl} \quad \frac{2 \times 0.0584}{0.965 + 0.0584} = 11.4\%$$

$$\text{H}_2 \quad \frac{0.535 - 0.0584}{0.965 + 0.0584} = 46.6\%$$

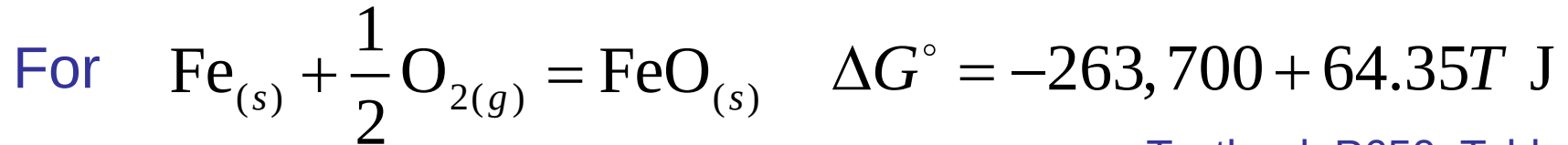
$$\text{Ar} \quad \frac{0.43}{0.965 + 0.0584} = 42\%$$

Chapter 12

12.6 Fe and FeO are in equilibrium with a gas mixture of composition 71.8% CO–28.2% CO₂ at 1273 K. Which of the two solid phases disappears if the composition of the gas is held constant and the temperature of the system is decreased?

Chapter 12

Solution to 12.6



Textbook P650, Table A.1

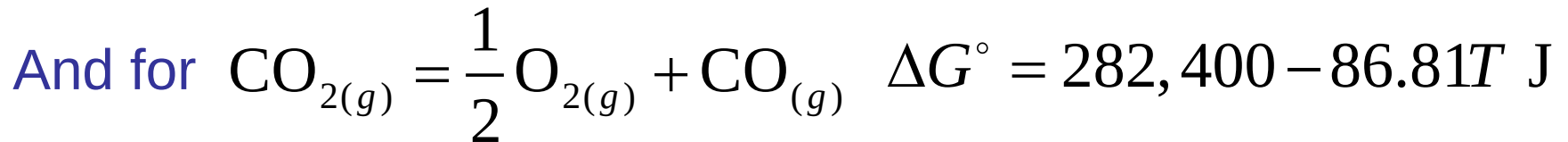


Table A.1 The Standard Gibbs Free Energy Changes for Several Reactions

Reaction	$\Delta G^\circ, \text{ J}$	Range, K
$2\text{Ag}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{Ag}_2\text{O}_{(s)}$	$-30,540 + 66.11 T$	298–463
$\text{Al}_{(l)} = [\text{Al}]_{(1 \text{ wt\% in Fe})}$	$-43,100 - 32.26 T$	
$2\text{Al}_{(l)} + 1.5\text{O}_{2(g)} = \text{Al}_2\text{O}_{3(s)}$	$-1,687,200 + 326.8 T$	993–2327
$\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{CO}_{(g)}$	$-111,700 - 87.65 T$	298–2000
$\text{C}_{(s)} + \text{O}_{2(g)} = \text{CO}_{2(g)}$	$-394,100 - 0.84 T$	298–2000
$\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} + \frac{1}{2}\text{S}_{2(g)} = \text{COS}_{(g)}$	$-202,800 - 9.96 T$	773–2000
$\text{C}_{(gr)} + 2\text{H}_{2(g)} = \text{CH}_{4(g)}$	$-91,040 + 110.7 T$	773–2000
$\text{C}_{(gr)} = [\text{C}]_{(1 \text{ wt\% in Fe})}$	$22,600 - 42.26 T$	

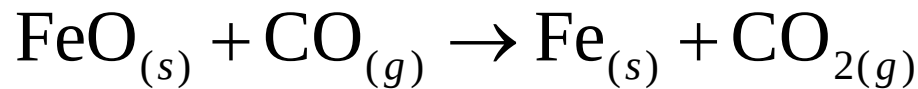
Chapter 12

Solution to 12.6

Adding gives $\text{Fe}_{(s)} + \text{CO}_{2(g)} = \text{FeO}_{(s)} + \text{CO}_{(g)}$

$$\Delta G^\circ = 18,700 - 22.46T \text{ J}$$

As written, the reaction is endothermic. Thus a decrease in temperature causes the equilibrium to shift as:



which decrease $p_{\text{CO}(\text{eq})}$ and increase $p_{\text{CO}_2(\text{eq})}$.

Thus decreasing the temperature at constant p_{CO} and constant p_{CO_2} cause the FeO phase to disappear.

Chapter 12

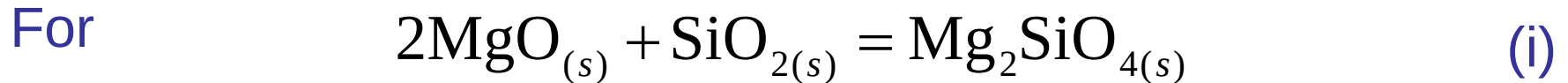
12.7 Calculate the vapor pressure of Mg exerted at 1400° C by the system in which the following reaction equilibrium is established.



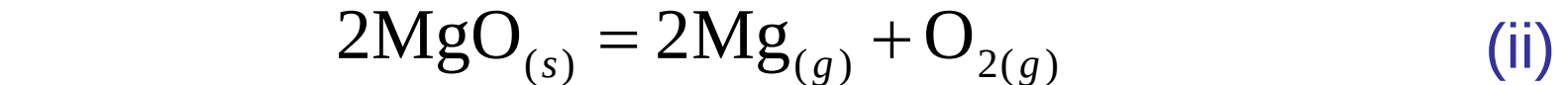
Chapter 12

Solution to 12.7

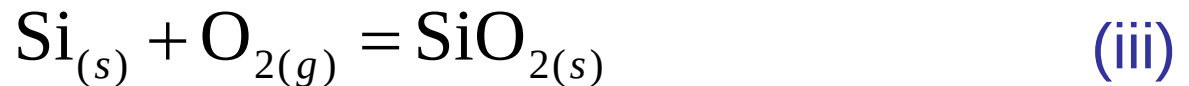
Textbook P650, Table A.1



$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= -67,200 + 4.31T \text{ J} \\ &= -67,200 + 4.31 \times 1673 \text{ J} = -59,989 \text{ J} \end{aligned}$$



$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= 2 \times (729,600 - 204T) \text{ J} \\ &= 2 \times (729,600 - 204 \times 1673) \text{ J} = 776,616 \text{ J} \end{aligned}$$

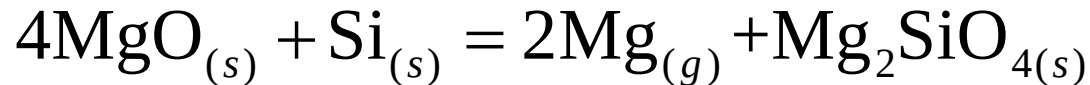


$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= -907,100 + 175T \text{ J} \\ &= -907,100 + 175 \times 1673 \text{ J} = -614,325 \text{ J} \end{aligned}$$

Chapter 12

Solution to 12.7

(i)+(ii)+(iii) gives



$$\Delta G_{1673\text{ K}}^{\circ} = 102,302\text{ J} = RT \ln \frac{1}{p_{\text{Mg}(\text{eq})}^2}$$

So

$$\begin{aligned} p_{\text{Mg}(\text{eq})} &= \exp\left(-\frac{\Delta G_{1673\text{ K}}^{\circ}}{2RT}\right) \\ &= \exp\left(-\frac{102,302}{2 \times 8.3144 \times 1673}\right) = 2.53 \times 10^{-2}\text{ atm} \end{aligned}$$

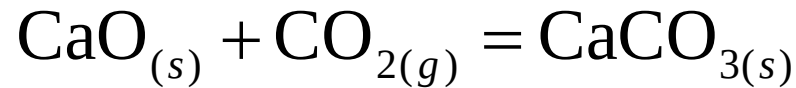
Chapter 12

12.8 One gram of CaCO_3 is placed in an evacuated rigid vessel of volume 1 liter at room temperature, and the system is heated. Calculate (a) the highest temperature at which the CaCO_3 phase is present, (b) the pressure in the vessel at 1000 K, and (c) the pressure in the vessel at 1500 K. The molecular weight of CaCO_3 is 100.

Chapter 12

Solution to 12.8

For



$$\Delta G^\circ = -168,400 + 144T \text{ J}$$

So

$$\ln p_{\text{CO}_2(\text{eq})} = \frac{\Delta G^\circ}{RT} = -\frac{168,400}{RT} + \frac{144}{R} = -\frac{20254}{T} + 17.32 \quad (\text{i})$$

1 g CaCO_3 , $n=1/100=0.01$ moles, so the vessel contains 0.01 moles of CO_2 . If all of the 0.01 moles of CO_2 occur in the gas phase, thus

$$p_{\text{CO}_2} = \frac{nRT}{V} = \frac{0.01 \times 0.082057T}{1} = 8.2057 \times 10^{-4} T \quad (\text{ii})$$

Chapter 12

Solution to 12.8

From (i)

$$p_{\text{CO}_2} = \exp \left[-\frac{20254}{T} + 17.32 \right] \quad (\text{iii})$$

(a) Lines (ii) and (iii) intersect at $p_{\text{CO}_2} = 0.96 \text{ atm}$, $T = 1173 \text{ K}$, which is the highest temperature at which CaCO_3 exists.

(b) At $T=1000 \text{ K}$, equation (iii) gives $p_{\text{CO}_2(\text{eq})} = 0.055 \text{ atm}$.

(b) At $T=1500 \text{ K}$, equation (ii) gives $p_{\text{CO}_2(\text{eq})} = 1.23 \text{ atm}$.

Chapter 12

12.9 Calculate the total pressure ($p_{\text{SO}_3} + p_{\text{SO}_2} + p_{\text{O}_2}$) exerted by equilibrated CoO and CoSO_4 at 1223 K.

Chapter 12

Solution to 12.9

For $\text{CoO}_{(s)} + \text{SO}_{3(g)} = \text{CoSO}_{4(s)}$ Textbook P650,
Table A.1

$$\Delta G_{1223 \text{ K}}^{\circ} = -227,860 + 165.3T \text{ J} = -25698 \text{ J}$$

$$\Delta G_{1223 \text{ K}}^{\circ} = RT \ln p_{\text{SO}_3(\text{eq})} = 8.3144 \times 1223 \ln p_{\text{SO}_3(\text{eq})}$$

Combining the two equations gives

$$p_{\text{SO}_3(\text{eq})} = 7.99 \times 10^{-2} \text{ atm} \quad (\text{i})$$

For $\text{SO}_{3(g)} = \text{SO}_{2(g)} + \frac{1}{2} \text{O}_{2(g)}$

$$\Delta G_{1223 \text{ K}}^{\circ} = 94,600 - 89.37T \text{ J} = -14699 \text{ J} = -8.3144 \times 1223 \ln \frac{p_{\text{SO}_2} p_{\text{O}_2}^{1/2}}{p_{\text{SO}_3}}$$

Chapter 12

Solution to 12.9

So

$$\frac{p_{\text{SO}_2} p_{\text{O}_2}^{1/2}}{p_{\text{SO}_3}} = 4.244 \quad \text{and} \quad p_{\text{O}_2} = \frac{1}{2} p_{\text{SO}_2}$$

Substitute (i)

$$4.244 = \frac{p_{\text{SO}_2} \cdot [0.5 p_{\text{SO}_2}]^{1/2}}{7.99 \times 10^{-2}}$$

Which gives

$$p_{\text{SO}_2} = 0.612 \text{ atm}, \quad p_{\text{O}_2} = 0.306 \text{ atm}, \quad p_{\text{SO}_3} = 0.0799 \text{ atm}$$

So

$$p_{\text{total}} = 1.0 \text{ atm}$$

Chapter 12

12.10 A gas mixture initially containing 90% CO, 0.4% COS, and 9.6% inert constituents (by volume) is passed over sponge iron at 1000 K to remove sulfur by the following reaction:

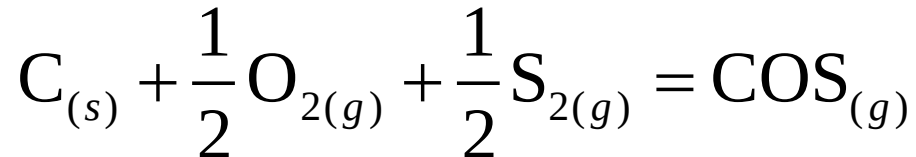


- Assuming that the effluent gas is in equilibrium with Fe and FeS, calculate the percentage of sulfur removed from the gas by reaction with the sponge iron.
- Calculate the partial pressure of S_2 in the effluent gas.

Chapter 12

Solution to 12.10

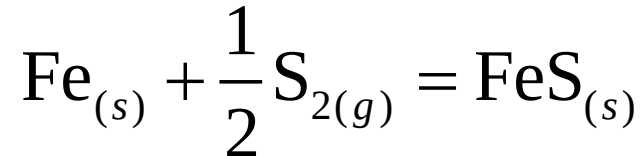
For



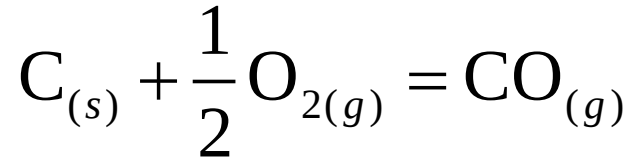
Textbook P650,

Table A.1

$$\Delta G_{1000 \text{ K}}^{\circ} = -202,800 - 9.96T \text{ J} = -212,760 \text{ J}$$

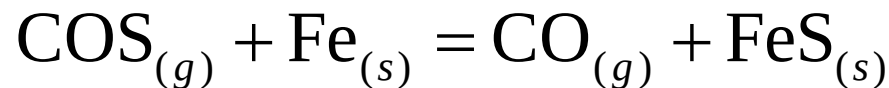


$$\Delta G_{1000 \text{ K}}^{\circ} = -150,200 + 52.55T \text{ J} = -97,650 \text{ J}$$



$$\Delta G_{1000 \text{ K}}^{\circ} = -111,700 - 87.65T \text{ J} = -199,350 \text{ J}$$

So



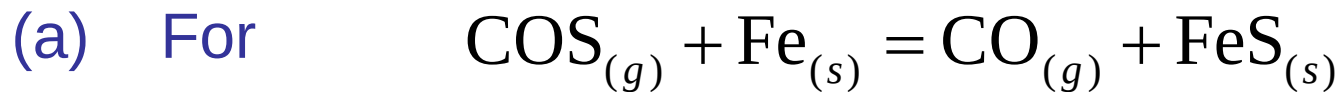
$$\Delta G_{1000 \text{ K}}^{\circ} = 212,760 - 97,650 - 199,350 \text{ J} = -84,240 \text{ J}$$

Chapter 12

Solution to 12.10

and

$$K_{1000\text{ K}} = \exp\left[\frac{-\Delta G_{1000\text{ K}}^{\circ}}{RT}\right] = \exp\left[\frac{84,240}{8.4144 \times 1000}\right] = 25130 = \frac{p_{\text{CO}}}{p_{\text{COS}}}$$



before 0.4 90

after 0.4-x 90+x

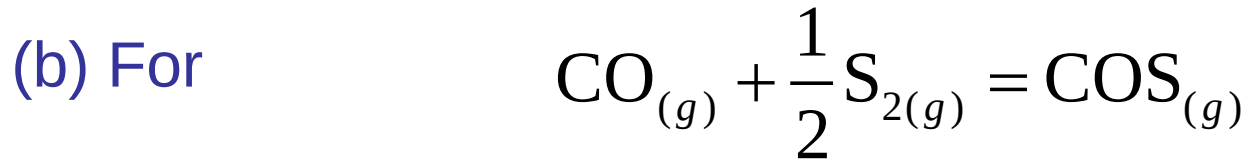
So
$$25130 = \frac{p_{\text{CO}}}{p_{\text{COS}}} = \frac{90+x}{0.4-x}$$

which gives $x=0.3964$.

Thus %S removed = $0.3964/0.4 \times 100\% = 99.1\%$

Chapter 12

Solution to 12.10



$$\Delta G_{1000 \text{ K}}^{\circ} = 199,350 - 212,760 \text{ J} = -13,410 \text{ J}$$

So

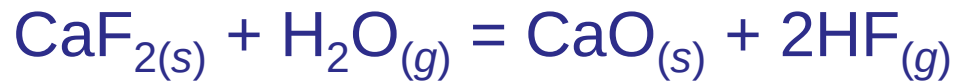
$$K_{1000 \text{ K}} = \exp \left[\frac{13410}{8.4144 \times 1000} \right] = 5.017 = \frac{p_{\text{COS}}}{p_{\text{CO}} p_{\text{S}_2}^{1/2}} = \frac{0.0036}{90.3964 p_{\text{S}_2}^{1/2}}$$

Which gives

$$p_{\text{S}_2} = 6.3 \times 10^{-11} \text{ atm}$$

Chapter 12

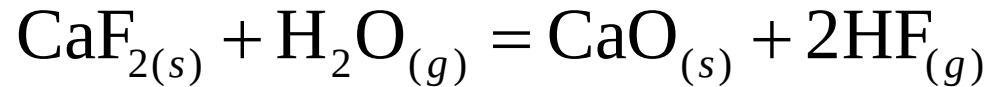
12.11 An Ar–H₂O gas mixture of $p_{\text{H}_2\text{O}} = 0.9 \text{ atm}$ ($P_{\text{total}} = 1 \text{ atm}$) is passed over solid CaF₂, as a result of which CaO forms according to



The reaction proceeds to equilibrium and solid CaO and solid CaF₂ are mutually immiscible. When the gas flow rate (measured at 298 K and 1 atm pressure) over the sample is 1 liter per minute, the measured rates of weight loss of the sample are 2.69×10^{-4} and 8.30×10^{-3} grams per hour at 900 and 1100 K, respectively. Use these data to calculate the variation of ΔG° for the preceding reaction with temperature. The atomic weights are O = 16, F = 19, and Ca = 40.08

Chapter 12

Solution to 12.11



MW	78.08	56.08
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So a weight loss of 22g produces 2 moles of HF and consumes 1 moles of H₂O, thus a weight loss of 1g products 0.0909 moles of HF and consumes 0.0455 moles of H₂O.

At 900 K for 1 hour,

weight loss of 2.69×10^{-4} g produces 2.44×10^{-5} moles of HF and consumes 1.22×10^{-5} moles of H₂O.

Chapter 12

Solution to 12.11

In 60 L (1 liter per minute) of gas in which $p_{\text{H}_2\text{O}} = 0.9 \text{ atm}$,

$$n_{\text{H}_2\text{O}} = \frac{PV}{RT} = \frac{0.9 \times 60}{0.082057 \times 298} = 2.208 \text{ moles}$$

So consumption of 1.22×10^{-5} moles of H_2O is insignificant.

$$p_{\text{HF}} = \frac{2.44 \times 10^{-5} \times 0.082057 \times 298}{60} = 9.94 \times 10^{-6} \text{ atm}$$

Thus

$$K_{900 \text{ K}} = \frac{p_{\text{HF}}^2}{p_{\text{H}_2\text{O}}} = \frac{(9.94 \times 10^{-6})^2}{0.9} = 1.09 \times 10^{-10}$$

and

$$\Delta G_{900 \text{ K}}^\circ = -8.3144 \times 900 \times \ln(1.09 \times 10^{-10}) \text{ J} = 171,596 \text{ J} \quad (\text{i})$$

Chapter 12

Solution to 12.11

Similarly, at 1100 K for 1 hour,

0.0083 g loss per hour produces 7.545×10^{-4} moles/hour of HF.

So

$$p_{\text{HF}} = \frac{7.545 \times 10^{-4} \times 0.082057 \times 298}{60} = 3.075 \times 10^{-4} \text{ atm}$$

$$K_{1100 \text{ K}} = \frac{(3.075 \times 10^{-4})^2}{0.9} = 1.051 \times 10^{-7}$$

$$\Delta G_{1100 \text{ K}}^{\circ} = -8.3144 \times 1100 \times \ln(1.051 \times 10^{-7}) \text{ J} = 146,962 \text{ J} \quad (\text{ii})$$

Chapter 12

Solution to 12.11

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ} = A + BT$$

Combines (i) and (ii),

$$171596 = A + 900B$$

$$146962 = A + 1100B$$

which gives,

$$B = -123 \text{ J/K}$$

$$A = 282449 \text{ J}$$

So

$$\Delta G^{\circ} = 282449 - 123T$$

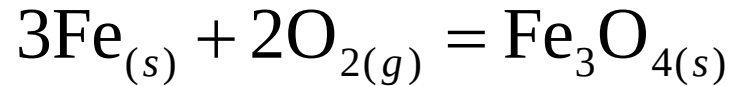
Chapter 12

12.12 Magnetite (Fe_3O_4) is reduced to sponge iron (Fe) in a continuous reactor operating at 800 K using methane gas (CH_4) as the reducing agent. The gaseous reaction product leaving the reactor at a total pressure of 1 atm is a mixture of CO, CO_2 , H_2 , and H_2O with a negligible methane content. The gas is at equilibrium with the Fe– Fe_3O_4 mixture in the reactor. Calculate the consumption of methane as moles of methane used per mole of sponge iron produced.

Chapter 12

Solution to 12.12

For



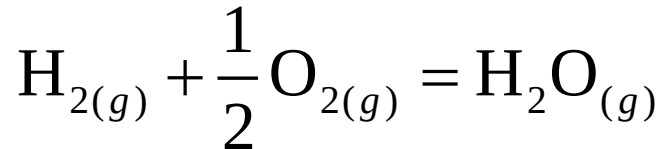
Textbook P650,
Table A.1

$$\begin{aligned}\Delta G_{800\text{ K}}^{\circ} &= -1,102,200 + 307.4T \text{ J} = -856,280 \text{ J} \\ &= RT \ln p_{\text{O}_2(\text{eq})}^2 = 2 \times 8.3144 \times 800 \ln p_{\text{O}_2(\text{eq})}\end{aligned}$$

So in the exit gas

$$p_{\text{O}_2(\text{eq})} = \exp\left[\frac{-856,280}{2 \times 8.3144 \times 800}\right] = 1.11 \times 10^{-28} \text{ atm}$$

For



$$\begin{aligned}\Delta G_{800\text{ K}}^{\circ} &= -247,500 + 55.85T \text{ J} = -202,560 \text{ J} \\ &= -8.3144 \times 800 \ln \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{1/2}}\end{aligned}$$

Chapter 12

Solution to 12.12

So
$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{1/2}} = 1.681 \times 10^{13} \quad \text{and} \quad \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.1771 \quad (\text{i})$$

For
$$2\text{CO}_{(g)} + \text{O}_{2(g)} = 2\text{CO}_{2(g)}$$
$$\Delta G_{800 \text{ K}}^{\circ} = 2(-282,400 + 86.81T) \text{ J} = -425,904 \text{ J}$$
$$= -2 \times 8.3144 \times 800 \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}} p_{\text{O}_2}^{1/2}}$$

So
$$\frac{p_{\text{CO}_2}}{p_{\text{CO}} p_{\text{O}_2}^{1/2}} = 8.02 \times 10^{13} \quad \text{and} \quad \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.845 \quad (\text{ii})$$

Chapter 12

Solution to 12.12

According to the title,

$$p_{\text{H}_2} + p_{\text{H}_2\text{O}} + p_{\text{CO}} + p_{\text{CO}_2} = 1 \text{ atm} \quad (\text{iii})$$

All of the C and H in the gas come from the CH_4 ,

so

$$\frac{n_{\text{C}}}{n_{\text{H}}} = \frac{1}{4} = \frac{p_{\text{CO}} + p_{\text{CO}_2}}{2p_{\text{H}_2} + 2p_{\text{H}_2\text{O}}}$$

$$p_{\text{H}_2} + p_{\text{H}_2\text{O}} = 2p_{\text{CO}} + 2p_{\text{CO}_2} \quad (\text{iv})$$

Solution of (i)-(iv) gives,

$$p_{\text{H}_2} = 0.5662 \text{ atm}, \quad p_{\text{H}_2\text{O}} = 0.1003 \text{ atm},$$

$$p_{\text{CO}} = 0.1806 \text{ atm}, \quad p_{\text{CO}_2} = 0.1528 \text{ atm}.$$

Chapter 12

Solution to 12.12

All of the C comes from CH_4

$$n_{\text{C}} = 0.1806 + 0.1528 = 0.3334 \text{ moles}$$

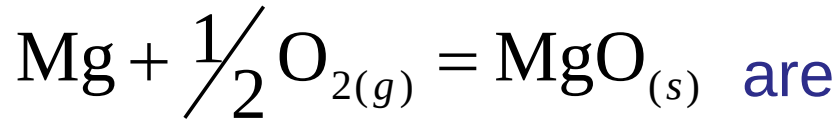
All of the O comes from Fe_3O_4

$$n_{\text{O}} = 0.1003 + 0.1806 + 2 \times 0.1528 = 0.5865 \text{ moles}$$

So 0.3334 moles of CH_4 per $3/4 \times 0.5865 = 0.4398$ moles of Fe,
or 0.76 moles of CH_4 per mole of Fe produced.

Chapter 12

12.13 Three equations for the oxidation of Mg according to



$$\Delta G^\circ = -604,000 - 5.36T \ln T + 142.0T \text{ J} \quad (\text{i})$$

$$\Delta G^\circ = -759,800 - 13.4T \ln T + 317T \text{ J} \quad (\text{ii})$$

$$\Delta G^\circ = -608,100 - 0.44T \ln T + 112.8T \text{ J} \quad (\text{iii})$$

One of these expressions is for the oxidation of solid Mg, one is for the oxidation of liquid Mg, and one is for the oxidation of gaseous Mg. Determine which equation is for which oxidation and calculate the melting and normal boiling temperature of Mg.

Chapter 12

Solution to 12.13

For (I)

$$\frac{\Delta G^\circ}{T} = \frac{-604,000}{T} - 5.36 \ln T + 142.0$$
$$\frac{\partial(\Delta G^\circ/T)}{\partial T} = \frac{604,000}{T^2} - \frac{5.36}{T} = -\frac{\Delta H^\circ}{T^2}$$

so

$$\Delta H_I^\circ = -604,000 + 5.36T$$

Similarly

$$\Delta H_{II}^\circ = -759,800 + 13.4T$$

$$\Delta H_{III}^\circ = -608,100 + 0.44T$$

Chapter 12

Solution to 12.13

Generally,

$$\left| \Delta H_{\text{Mg}(g)}^{\circ} \right| > \left| \Delta H_{\text{Mg}(l)}^{\circ} \right| > \left| \Delta H_{\text{Mg}(s)}^{\circ} \right|$$

So equation (II) is for oxidation of $\text{Mg}_{(g)}$, equation (III) is for oxidation of $\text{Mg}_{(l)}$, equation (I) is for oxidation of $\text{Mg}_{(s)}$.

$$\Delta G_{\text{I}}^{\circ} = \Delta G_{\text{III}}^{\circ} \quad \text{at} \quad T_{m,\text{Mg}} = 930 \text{ K}$$

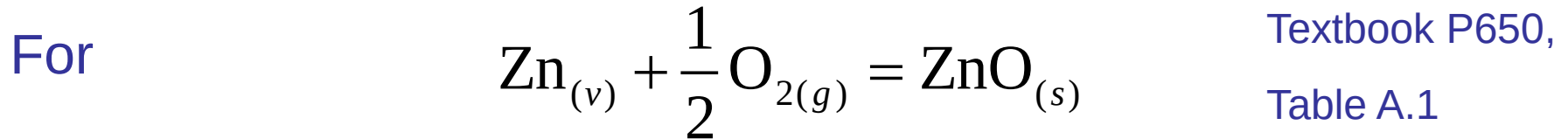
$$\Delta G_{\text{II}}^{\circ} = \Delta G_{\text{III}}^{\circ} \quad \text{at} \quad T_{b,\text{Mg}} = 1372 \text{ K}$$

Chapter 12

12.14 Two hundred grams of liquid zinc are placed in a crucible at 1030 K. Two moles of air are bubbled through the liquid zinc, and the gas comes to equilibrium with the liquid before leaving the system. If the total pressure of the gas remains constant at 0.8 atm throughout the process, how many grams of metallic zinc are left in the crucible? The atomic weights of Zn and O are, respectively, 65.38 and 16.

Chapter 12

Solution to 12.14



$$\Delta G_{1030 \text{ K}}^{\circ} = -460,200 + 198T \text{ J} = -256,260 \text{ J} = -RT \ln K_{1030 \text{ K}}$$

So
$$K_{1030 \text{ K}} = 9.9 \times 10^{12} = \frac{1}{p_{\text{Zn}} p_{\text{O}_2}^{1/2}}$$

$$\ln p_{\text{Zn}}^{\circ} (\text{atm}) = -\frac{15250}{T} - 1.255 \ln T + 21.79$$
 Textbook P654, Table A.4

Thus at 1030 K,
$$p_{\text{Zn}}^{\circ} = 0.1787 \text{ atm}$$

$$p_{\text{O}_2} = \left[\frac{1}{9.9 \times 10^{12} \times 0.1787} \right]^2 = 3.2 \times 10^{-25} \text{ atm}$$

Chapter 12

Solution to 12.14

2 moles air: 0.42 moles O₂ + 1.58 moles N₂

0.42 moles O₂ + 0.84 moles Zn produce 0.84 moles ZnO

and the weight of 0.84 moles ZnO is: 0.84 x 65.38 = 54.92 g

So 54.92 g of Zn are oxidized to ZnO

1.58 moles of N₂ at 1030 K, $p_{\text{total}} = 0.8 \text{ atm}$

So $p_{\text{N}_2} = 0.8 - 0.1787 = 0.621 \text{ atm}$

$$p_{\text{N}_2} V = n_{\text{N}_2} RT$$

Which gives the volume of gas

$$V = \frac{1.58 \times 0.082057 \times 1030}{0.621} = 215 \text{ L}$$

Chapter 12

Solution to 12.14

$$p_{\text{Zn}}V = n_{\text{Zn}}RT$$

So

$$n_{\text{Zn}} = \frac{0.1787 \times 215}{0.082057 \times 1030} = 0.455 \text{ moles}$$

The weight of 0.455 moles Zn is: $0.455 \times 65.38 = 29.78 \text{ g}$

Thus 29.78 g Zn are evaporated, and the remaining liquid Zn is:

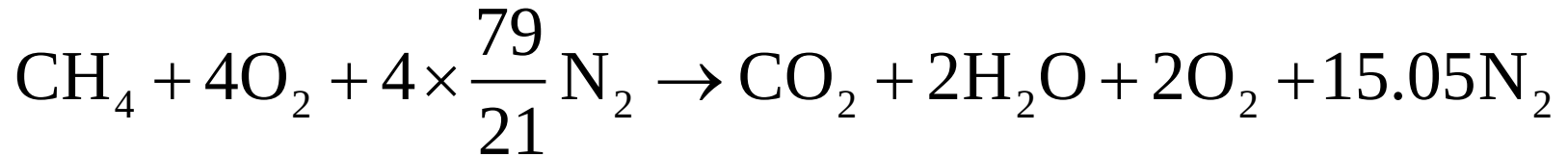
$$200 - 54.9 - 29.8 = 115.3 \text{ g}$$

Chapter 12

12.15 Methane gas is burned with twice the amount of stoichiometric air (i.e., the initial ratio of CH_4 and O_2 in the mixture is 0.25), and the combustion gas produced, in which the concentration of methane is negligible, is used to calcine CaCO_3 in an isothermal furnace. The gas and the solids are at 1080 K and the pressure of the gas is maintained constant at 1 atm. How many moles of CaCO_3 are decomposed per mole of CH_4 burned?

Chapter 12

Solution to 12.15

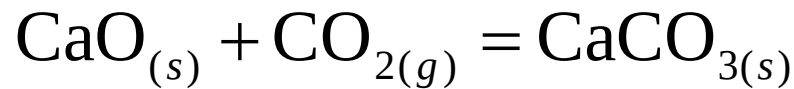


$$n_{\text{total}} = 20.05 \text{ moles}$$

In this gas

$$p_{\text{CO}_2} = \frac{1}{20.05} = 0.0499 \text{ atm}$$

For



Textbook P650,
Table A.1

$$\begin{aligned} \Delta G_{1080 \text{ K}}^{\circ} &= -168,400 + 144T \text{ J} = -13124 \text{ J} \\ &= 8.3144 \times 1080 \ln p_{\text{CO}_2(\text{eq})} \end{aligned}$$

Which gives

$$p_{\text{CO}_2(\text{eq})} = 0.232 \text{ atm}$$

Chapter 12

Solution to 12.15

The decomposition of x moles of CaCO_3 produces x moles of CO_2 and increases p_{CO_2} from 0.0499 atm to 0.232 atm.

That is

$$0.232 = \frac{1 + x}{20.05 + x}$$

So $x = 4.76$

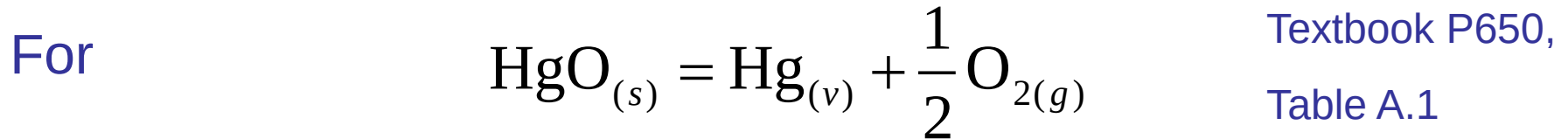
And thus 4.76 moles of CaCO_3 are decomposed per moles of CH_4 burned.

Chapter 12

12.16 Mercuric oxide (HgO) is placed in a vessel which is then evacuated, filled with nitrogen, and heated to 600 K, at which temperature it is observed that the total pressure in the vessel is 2 atm. Calculate the mole fractions of O_2 and Hg vapor in the gas phase.

Chapter 12

Solution to 12.16



$$\Delta G_{600\text{ K}}^{\circ} = 152,200 - 207.2T \text{ J} = 27,880 \text{ J}$$

So $K_{600\text{ K}} = \exp\left[\frac{-\Delta G_{600\text{ K}}^{\circ}}{RT}\right] = 3.74 \times 10^{-3} = p_{\text{Hg}} p_{\text{O}_2}^{1/2}$

Obviously $p_{\text{O}_2} = \frac{1}{2} p_{\text{Hg}}$, so

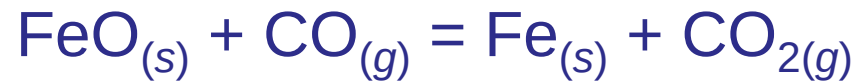
$$3.74 \times 10^{-3} = p_{\text{Hg}} [0.5 p_{\text{Hg}}]^{1/2}$$

Which gives $p_{\text{Hg}} = 0.0304 \text{ atm}$, $p_{\text{O}_2} = 0.0152 \text{ atm}$, $p_{\text{N}_2} = 1.954 \text{ atm}$

Thus $X_{\text{Hg}} = \frac{0.0304}{2} = 0.0152$, $X_{\text{O}_2} = \frac{0.0152}{2} = 0.0071$

Chapter 12

12.17 In Figure 12.17 the line AB, which represents the equilibrium



intersects the carbon deposition line at the point A

($T = 927 \text{ K}$, $p_{\text{CO}_{(s)}} = 0.595 \text{ atm}$, $p_{\text{CO}_2} = 0.405 \text{ atm}$, $P = 1.000 \text{ atm}$).

At what total pressure does the point of intersection of the lines occur at 1000 K, and what are the values of the partial pressures of CO and CO₂ in this state?

Chapter 12

Solution to 12.17

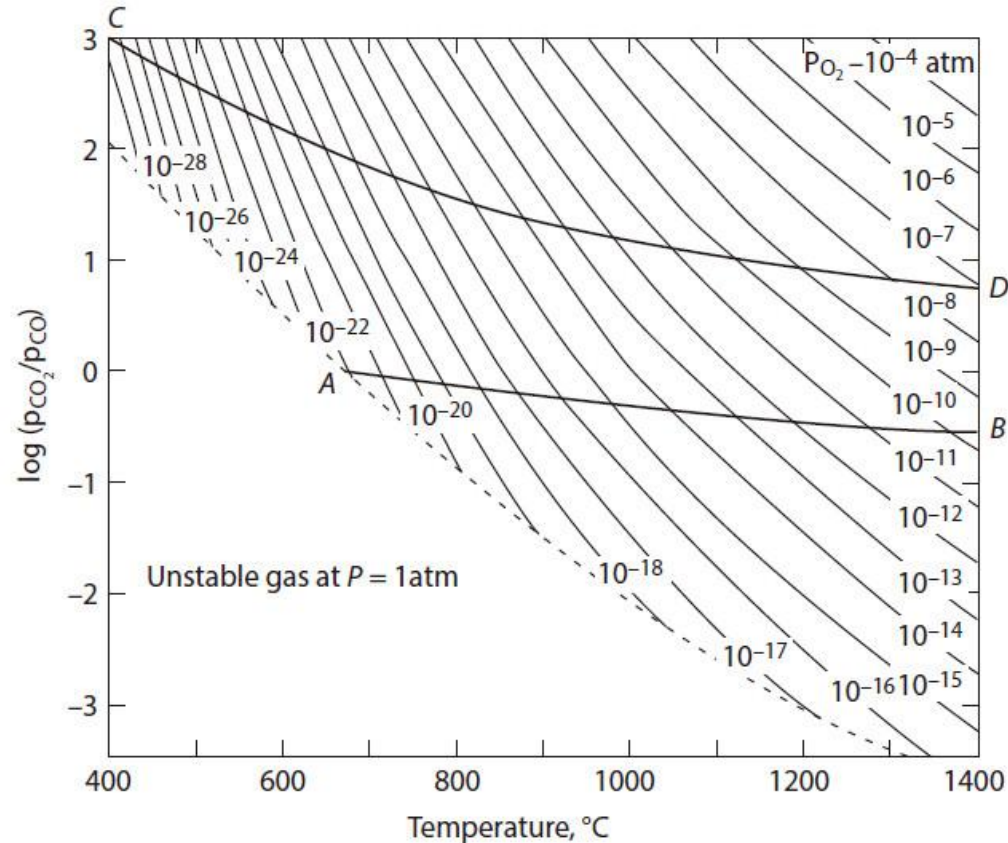
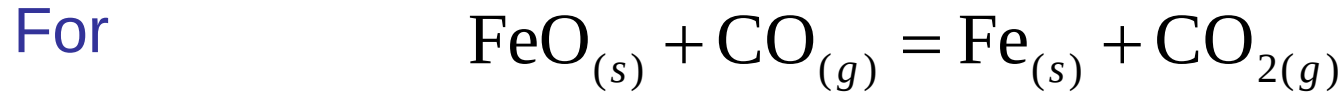


Figure 12.17 The relationship between the partial pressure of oxygen in a CO_2 – CO gas mixture and temperature. The broken line is the variation, with temperature, of the composition of the gas which is in equilibrium with graphite at 1 atm pressure. The lines AB and CD represent, respectively, the equilibria $\text{Fe} + \text{CO}_2 = \text{FeO} + \text{CO}$ and $\text{Co} + \text{CO}_2 = \text{CoO} + \text{CO}$.

Chapter 12

Solution to 12.17



$$\Delta G^\circ = -22,800 + 24.26T \text{ J} = -RT \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}}$$

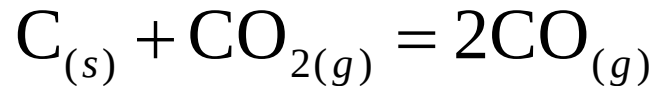
So at the total pressure P_T

$$\frac{2249}{T} - 2.70 = \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = \ln \frac{P_T - p_{\text{CO}}}{p_{\text{CO}}} \quad (\text{i})$$

Chapter 12

Solution to 12.17

For



$$\Delta G^\circ = 170,700 - 174.46T \text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2}}$$

So at the total pressure P_T

$$-\frac{20,530}{T} + 20.99 = \ln \frac{p_{\text{CO}}^2}{P_T - p_{\text{CO}}} \quad (\text{ii})$$

With $T=1000 \text{ K}$, lines (i) and (ii) gives

$$P_T = 1.651 \text{ atm, and } p_{\text{CO}} = 1.009 \text{ atm, } p_{\text{CO}_2} = 0.642 \text{ atm}$$

Thermodynamics of Materials & Phase transformations

The End

